

Recall

- Schauder basis
- reln. between Hamel & Schauder
- Separability
- Finite dimensional NLS $\Rightarrow$ separability.

Exercise.

• Let X be a nls and $Y \subseteq X$ dense.

If Y is separable $\Rightarrow X$ is separable.

(Hint $\overline{Y} = X$ $D \subseteq Y$ ^{countable})

$$\overline{D^Y} = Y \quad \overline{D^Y} = Y \cap \overline{D^X}$$

Claim: $\overline{D^X} = X$,

Theo. A NLS with Schauder basis is separable.

Proof: Let X be a NLS with the
Schauder basis
 $\{u_1, u_2, \dots\}$.

Let $Y_n = \text{span}\{u_1, u_2, \dots, u_n\}$.

Define $Y = \bigcup_{n=1}^{\infty} Y_n$.

$\therefore Y$ is dense in X . (Schauder basis).

It is enough for us to
find a countable dense subset
of Y .

Consider $D_n = \left\{ \sum_{j=1}^n \alpha_j u_j \mid \alpha_j \in \mathbb{Q}_k \right\}$

Then D_n is a countable dense

set in Y_n .

Consider $D = \bigcup_{i=1}^{\infty} D_i$ is a countable dense in Y .

So from $\Rightarrow$ the above ex
 X is separable.

Ex.

1) C_{00} , $\mathcal{P}[a, b]$ endowed with $\|\cdot\|_{\infty}$ norm are separable.

2) l^p is separable.

3.) $(C[a, b], \|\cdot\|_{\infty})$ is separable
since $\mathcal{P}[a, b]$ is separable.

4) $C[a, b]$ is dense in $L^p[a, b] \Rightarrow L^p[a, b]$ is separable.

• l^∞ is NOT separable.

Proposition.

$(\mathcal{U}, d)$ be a m.s. If there exists an uncountable collection of disjoint open sets then $\mathcal{U}$ is not separable.

In particular; if $\exists$ an uncountable set E and $r > 0$ s.t
 $d(x, y) \geq r \quad \forall x, y \in E, x \neq y$
then $\mathcal{U}$ is not separable.

Proof: $\mathcal{G} \Rightarrow$ uncountable collection
of disjoint open
sets in Ω .

Let $D \subseteq \Omega$ dense.

$\exists x_G \in D \cap G$ for each $G \in \mathcal{G}$.

Since $\mathcal{G}$ is uncountable so

$\{x_G \mid G \in \mathcal{G}\}$ is uncountable

$\subseteq D$.

$\therefore D$ is uncountable.

$\therefore \Omega$ is not separable.

ℓ^∞ is NOT separable.

$$E = \left\{ x \in l^\infty \mid x(j) = \{0,1\} \right\}.$$

E is uncountable.

$$\forall x, y \in E \quad x \neq y$$

$$\|x - y\|_\infty = 1.$$

$\therefore l^\infty$ is not separable.

• $L^\infty[a, b]$

$$E = \left\{ \chi_{[a, s]} \mid a \leq s \leq b \right\}$$

$$\|x - y\|_\infty = 1 \quad \forall x, y \in E$$

Schauder basis $\Rightarrow$ separability

$\Leftarrow$?
Negative Enflo.

19.7.3

Recall

Let (X, d) be a M.S

1) (X, d) is compact

2) (X, d) is seq compact.

Every sequence has a convergent subsequence.

3) (X, d) is complete
& totally bdd.

(1), (2), (3) are equivalent.

⇒ A compact subset of a
m.s $\Rightarrow$ closed & bdd.

→ In $\mathbb{K}^n$, with euclidean norm
closed + bdd $\Leftrightarrow$ compact.

Lemma Let X and Y be NLS
say $\|\cdot\|_X$ & $\|\cdot\|_Y$ resp.

Suppose $T: X \rightarrow Y$ is a
bijective linear transformation

s.t $\exists c_1, c_2 > 0$

satisfying

$$\underline{c_1 \|x\|_X \leq \|T(x)\|_Y} \leq \underline{c_2 \|x\|_X}.$$

Then ,

(1) X is complete $\Leftrightarrow Y$ is complete

(2) $S \subseteq X$ is closed (bdd)

$\Leftrightarrow T(S) \subseteq Y$ is closed (bdd).

(3) $S \subseteq X$ is compact

$\Leftrightarrow T(S) \subseteq Y$ is compact

Proof.

(x_n) is Cauchy in X

$\Leftrightarrow y_n = T(x_n)$ is Cauchy in Y .

Heine - Borel theo.

Theo. Every closed & bdd
subset of a Finite-dimensional
NLS is compact.

Proof: Let $(X, \|\cdot\|)$ be
the given F.d NLS.

$$\bullet E = \{u_1, \dots, u_n\}$$

Recall

$$\xrightarrow{\text{norm}} \|x\|_E = \max \left\{ |f_1(x)|, \dots, |f_n(x)| \right\}$$

$$\left(\begin{array}{l} x = \alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n \\ f_j : X \rightarrow \mathbb{K} \end{array} \right),$$

$$f_j(x) = \alpha_j \quad j=1, 2, \dots, n$$

$$T: X \longrightarrow \mathbb{K}^n$$

$$\begin{aligned} T(x) &= T(\alpha_1 u_1 + \dots + \alpha_n u_n) \\ &= (\alpha_1, \alpha_2, \dots, \alpha_n) \end{aligned}$$

- T is bijective linear transformation.

$$\begin{aligned} \|T(x)\|_\infty &= \max \{ |\alpha_1|, \dots, |\alpha_n| \} \\ &= \|x\|_E \end{aligned}$$

Since X is finite dim.
So $\|\cdot\|$ & $\|\cdot\|_E$ are
equivalent.

$\exists c_1, c_2 > 0$ s.t

$$c_1 \|x\| \leq \|x\|_E \leq c_2 \|x\|$$

$$c_1 \|x\| \leq \|T(x)\|_\infty \leq c_2 \|x\|$$

— ①

$\therefore$ By Lemma.

$S \subseteq X$ is closed & bdd
in X

$\Rightarrow T(S)$ is cl & bdd

$\Rightarrow T(S)$ is cl & bdd
 in $\mathbb{K}^n$ wrt $\|\cdot\|_\infty$
 wrt Euclidean norm

$\Rightarrow T(S)$ is compact
 in $\mathbb{K}^n$ wrt EN.

$\Rightarrow S$ is compact in X

Remark

Finite dimensionality
 assumption be dropped.

$$X = C_{\infty}, \quad \|\cdot\|_{\infty}.$$

$$S = \{x \in X \mid \|x\| = 1\}$$

$$(e_n)_{n \in \mathbb{N}} \quad \|e_n\|_{\infty} = 1$$

$$e_n \in S$$

$$\|e_n - e_m\|_{\infty} = 1 \quad \forall n \neq m$$

$\therefore (e_n)$ does not have
a Cauchy subseq

$\therefore S$ is not totally bdd.

'Every closed & bdd set is compact' is a characterization of finite dim.

Riesz Lemma

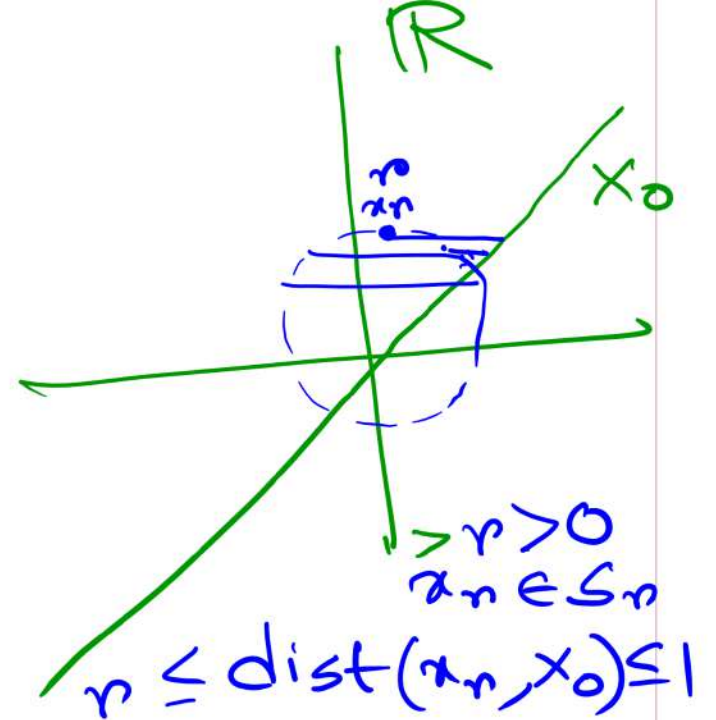
Let X_0 be a proper subspace of a NLS X .

Then for any r ,
 $0 < r < 1$

$\exists x_r \in X$ st $\|x_r\| = 1$

$\& \quad r \leq \text{dist}(x_r, X_0) \leq 1$ (Proof later)

Theo. X be a
NLS.



1) X is finite dim.

2) Every closed bdd subset of X is compact

3.) $D = \{x \in X \mid \|x\| \leq 1\}$ is compact.

4) For some $x_0 \in X$ & $r > 0$, $B(x_0, r)$ is totally bdd.

(1), (2), (3), (4) are equivalent

(3) $\Rightarrow$ (4).

Let $x_0 \in X$ & $r > 0$

Let (x_n) be a seq in
 $B(x_0, r)$

We have $\|x_n - x_0\| < r$.

Consider

$$y_n = \frac{x_n - x_0}{r}$$

$$\|y_n\| < 1$$

$\Rightarrow y_n \in D$. and (y_n) is

a seq in D .

Given D is compact so,

$\exists$ a seq $y_{n_j} \rightarrow y$ in D

Look at

$$x_{n_j} = r y_{n_j} + x_0$$



$$r y + x_0$$

Since (x_{n_j}) is convergent

$\Rightarrow (x_{n_j})$ is Cauchy.

$\therefore B(r_0, r)$ is totally
bdd.